

Multi-Surface Patient Ad Targeting with EMR/PMS Attribution

1. Field

Targeted media delivery in healthcare, personal-services, and point-of-care settings. Systems that select content for multiple delivery surfaces (in-room displays, lobby and waiting-room displays, checkout displays, and patient mobile devices) using (a) the venue's EMR/PMS data identifying each scheduled patient, (b) the practice's stated sales objectives, and (c) closed-loop attribution that ties each ad exposure on any surface back to subsequent transactions recorded in the same EMR/PMS.

2. Background

- **In-office signage** (PatientPoint, Health Monitor, Outcome Health) plays generic content to whoever is in the room. No patient identity, no per-patient targeting, no link to what the practice is trying to sell.
- **Patient-engagement platforms** (Phreesia, Klara, Relatient) personalize tablet content at check-in based on patient identity, but only at one surface (the tablet) and only at one moment (check-in). They do not coordinate with mobile or with revenue priorities.
- **Patient marketing platforms** (HubSpot, Mailchimp, Klaviyo, healthcare SMS) can reach known patients on mobile but have no awareness of clinic location, real-time schedule, or which screen the patient is standing in front of.
- **DOOH and POS-data signage** (Vistar Media, Vibonomics, Cooler Screens) optimize spend across screens or modulate content from aggregate sales data. No patient identity, no clinic schedule, no mobile coordination.

No existing system treats the clinic as a coordinated multi-surface delivery network where (a) patient identity and location are known from EMR/PMS data, (b) the practice has declared in machine-readable form what it wants to sell, and (c) content selection across every surface the patient encounters is driven by reconciling those two inputs and measured back against EMR/PMS-recorded transactions.

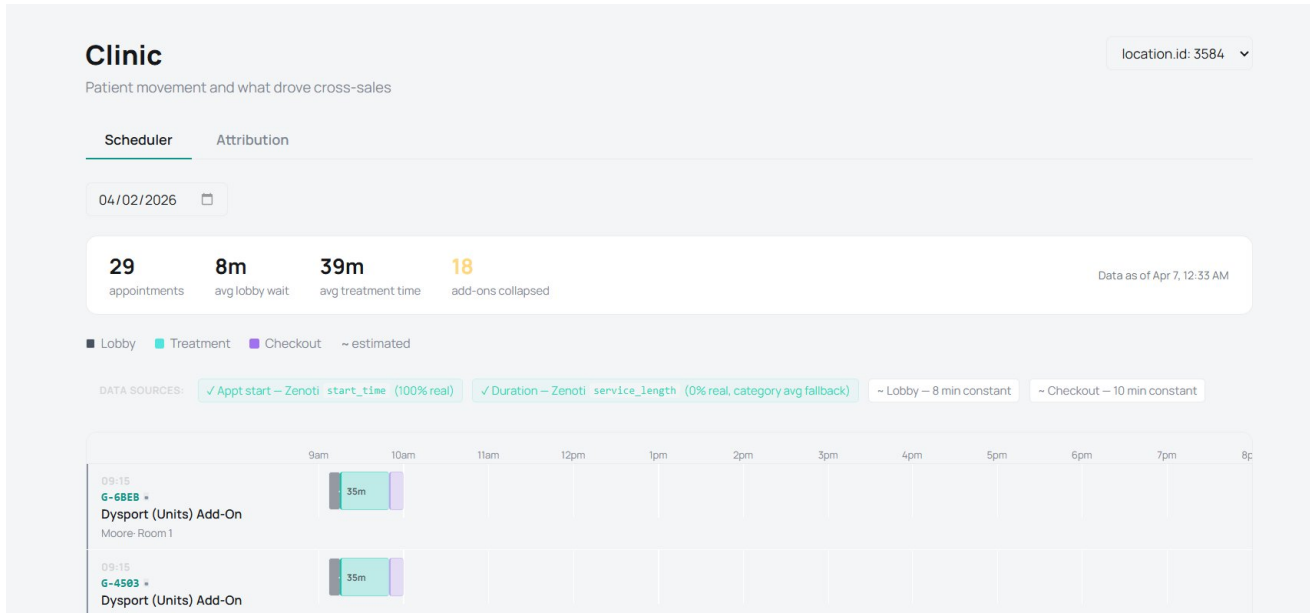
The invention gives clinics a wedge to drive their own sales goals through a closed-loop attribution system. The practice declares goals (e.g. "grow Morpheus8 by 30% this quarter," "fill the Tuesday Botox slow window," "reduce churn in 90-day-lapsed VIPs"). The EMR/PMS provides who is scheduled, what they have purchased, who they are. The system selects content for each surface the patient will encounter (pre-visit mobile, lobby display on arrival, treatment-room display pre and post procedure, checkout display, post-visit mobile), and every exposure is attributed back to any resulting transaction in the EMR/PMS.

3. Summary

- **Goal layer.** Clinic operator declares revenue objectives in machine-readable form (service to grow, segment to retain, time window to fill, target lift, weight). Time-bounded and audit-logged.
- **EMR/PMS pipeline.** Backend service that pulls appointment, transaction, and guest data; derives per-patient profile (LTV segment, return rate, service history, next-best-action); maintains a real-time map of who is scheduled where and when.
- **Surface-aware selector.** For each (patient, surface, time-window) tuple, the system selects content by reconciling the patient's next-best-action with active goals. Surfaces include in-room displays, lobby and waiting displays, checkout displays, and the patient's mobile device (SMS, push, in-app). Handles surface-specific constraints: 1:1 in-room vs multi-viewer lobby, opt-in mobile vs ambient display.

- **Spatial-temporal scheduling.** The clinic is modeled as a graph of screens, each registered with a zone type (treatment room, lobby, checkout, hallway) and a room or location ID. The EMR/PMS appointment binds each patient to specific rooms at specific times. The system produces a per-patient journey of touchpoints across surfaces with content selected for each.
- **Closed-loop attribution.** Every exposure of an identified patient to a content slot on any surface is logged, joined to subsequent EMR/PMS transactions, and classified as converted, in-window, or expired. Attribution feeds back into the recommendation model and the goal-pacing logic, so the system learns which surface and sequence drives conversion for which service and segment.

Reference screens



Scheduler. Per-clinic appointment timeline with lobby, treatment, and checkout windows by room. Sourced from EMR/PMS appointment data; binds patients to specific rooms at specific times for the spatial-temporal scheduling component.

Clinic

Patient movement and what drove cross-sales

location.id: 3584

Scheduler

Attribution

30d 90d 180d Data as of Apr 7, 12:33 AM

28 of 30 cross-sales had Vrtly exposure before conversion

93% exposure rate

PATIENT	SEGMENT	CROSS-SELL PATH	CONVERTED	VISIT DUR.	LOBBY	ROOM	INFO PACK	TOTAL
G-BB9E	Occasional	Dysport *3 → Profound RF Microneedling	Apr 3	45m	*6	*3	—	9
G-97E5	Ghost	Dysport *11 → HRT Lab Draw Fee	Apr 3	60m	*6	*6	✓	13
G-8BF9	Occasional	Signature Hydrafacial *4 → Dysport	Apr 3	45m	*5	*2	✓	8
G-2474	Occasional	Lesion Removal 15 Min *4 → Dysport	Apr 3	45m	*5	*2	—	7
G-7E3B	VIP	GLP-1 + BPC157 Injection *16 → Daxify	Apr 3	45m	*6	—	✓	7
G-DE7B	Occasional	Signature Hydrafacial *2 → Botox	Apr 3	45m	*8	*5	✓	14

Attribution. Cross-sales joined to prior Vrtly exposures across lobby, room, and Info Pack surfaces. Each row is an identified patient, the cross-sell path, exposure counts per surface, and the converting visit date drawn from EMR/PMS transaction records.

Guest Journey Intelligence

Cross-sell paths — distinct treatment categories only

location.id: 3584 location.id: 3585 Both 90d 180d 365d



Injectables → Skin · 96 guests · avg \$1,063 2nd visit

Injectable patients often explore skin treatments next. Show HydraFacial or Microneedling content to Injectables patients.

Category Cross-Sell Paths

Which treatment categories lead to the next booking — by volume

Injectables

96 guests

→

Skin

avg \$1,063 2nd visit · 40d between

Dysport → Signature Hydrafacial (14 guests)
 Dysport → Microneedling - 1 Area (10 guests)
 Dysport → Microneedling 1 Area - Series of 3 (5 guests)

Show HydraFacial/Microneedling to injectable patients

Skin

86 guests

→

Injectables

avg \$2,016 2nd visit · 47d between

Signature Hydrafacial → Dysport (10 guests)
 Microneedling - 1 Area → Dysport (8 guests)
 PRF Stand-Alone Service → Dysport (5 guests)

Introduce neuromodulators to skincare patients

Injectables

85 guests

→

Medical

avg \$526 2nd visit · 36d between

Dysport → HRT Pellet Insertion Female (26 guests)
 Dysport → Existing Client Consultation (6 guests)

Promote HRT consultations to long-term injectable patients

Goals and cross-sell paths. Category-level cross-sell volumes, average second-visit value, and time between visits. Used to seed practice sales objectives and to map detected goals to content categories shown across surfaces.

4. Embodiment

- Clinic registers each display in a device fleet layer with zone type and room or location ID.
- Operator sets sales goals through a structured interface (service, target, window, weight).
- EMR/PMS connector (initial: Zenoti REST API; architecture is EMR/PMS-agnostic) hydrates appointment cache, guest profile cache, and transaction history.
- Patient state machine tracks scheduled, checked-in, in-treatment, post-treatment, departed using EMR/PMS status and time inference.
- Selector runs per (surface, time-window) and outputs a content ID drawn from a pre-approved library, with a logged rationale tying the selection to patient profile and active goals.

- Mobile delivery via existing channels (SMS provider, push, in-app) keyed to EMR/PMS-known contact records, respecting recorded consent and opt-in.
- Attribution writer joins the cross-surface exposure log to the EMR/PMS transaction log on patient ID and time window.

Foundation already live: Zenoti integration, MySQL guest profile cache, six-detector analyst layer, per-slot hourly media plan, attribution tracking. Active at two centers (~5,000 cached appointments, ~\$1.3M revenue under management). The spatial display graph, the goal-input layer, and the mobile surface are the extensions covered by this brief.

5. What We Want to Protect

- Selecting media for any surface a patient will encounter (in-room display, lobby display, checkout display, personal mobile device) using EMR/PMS-derived identity, schedule, and history of that specific patient.
- Reconciling per-patient relevance with declared, machine-readable practice sales objectives to drive content selection on any of those surfaces.
- Binding content slots on physical displays to specific rooms at specific times based on appointment data, including the treatment room itself during the appointment window.
- Coordinating cross-surface touchpoints (pre-visit, in-clinic, post-visit) for an identified patient as a single planned journey.
- Closed-loop attribution that logs exposures across all surfaces, joins them to EMR/PMS transactions, and feeds outcomes back into the recommendation model and the goal-pacing logic.
- The combined system as a single revenue and media layer across the clinic's physical and mobile surfaces.

6. Prior Art

- **In-clinic signage** (PatientPoint, Health Monitor, Outcome Health): no patient identity, no goal layer, single-surface, no attribution back to EMR/PMS transactions.
- **Patient-engagement tablets** (Phreesia, Klara, Relatient): identified patient, but single surface, single moment, no goal layer, no in-room or checkout or mobile coordination.
- **Mobile patient marketing** (HubSpot, Mailchimp, Klaviyo, healthcare SMS): mobile only, no physical-space coordination, no real-time clinic location awareness.
- **DOOH and POS-data signage** (Vistar Media, Vibonomics, Cooler Screens): no patient identity, no clinic schedule, no mobile coordination, no EMR/PMS attribution.

No known system combines identified-patient targeting, declared sales objectives, multi-surface delivery across physical and mobile, and EMR/PMS attribution in one system. Counsel to validate with formal search.

7. Compliance

Mixed-practice settings may include HIPAA-covered information in the EMR/PMS. The system: (a) does not display patient names or conditions on screen, (b) keeps EMR/PMS-derived signals server-side and selects only from a pre-approved content library, (c) respects existing patient consent and opt-in records for mobile delivery, and (d) lets operators audit and override every mapping. Counsel to advise whether to position audit and consent as claim elements or as background.